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ABSTRACT

A new numerical simulation system has been developed which predicts flow behavior of fuel film formed on intake port and combustion chamber walls of gasoline engines. The system consists of a film flow model employing film thickness as a dependent variable, an air flow model, and a fuel spray model. The system can analyze fuel film flow formed on any arbitrary three-dimensional configuration. Fuel film flow formed under a condition of continuous intermittent fuel injection and steady-state air flow was calculated, and comparison with experimental data showed the system possessing ability of qualitative prediction.

INTRODUCTION

Cleaner emission and better fuel economy are main environmental issues automobile manufacturers must overcome. Control of combustion in engine cylinder is a key technique to solve the problems, and to achieve that, control of air and / or fuel is necessary.

Numerical simulation is one of the main approach to those studies, which has advantages over experimental approach in the stand point of better understanding of the phenomena taking place and easier access to alternative designs.

Numerical simulation of air flow in internal combustion engine is widely studied both in gasoline and diesel engines employing various kind of techniques (1,2). In diesel engines, fuel behavior is simulated by discrete droplet model (DDM), by which fuel flying in air as well as after impingement on combustion chamber wall are studied (2,3). In gasoline engines, injected fuel in intake port is also modeled by DDM approach (4), however, fuel behavior after attachment on the wall has not been modeled. The fuel film flow on intake port and combustion chamber wall is an important phenomena relating to emission and fuel economy as well as acceleration response or starting up performance especially in cold engine condition.

Here, film flow model is newly developed, and combined with a spray model and an air flow model to build a system which can analyze fuel behavior in intake port and combustion chamber of gasoline engines.

NUMERICAL MODELS AND THE SYSTEM

In multi-port-injection gasoline engines, fuel is injected at entrance of intake port in form of spray. The spray motion can be affected by air flow in the intake port, so that some part of the spray flies into the cylinder and some arrives on to the port wall or the intake valve. The fuel arriving on the solid surface forms film flow, and flows into combustion chamber, most of them are in liquid state.

Therefore, three models, a fuel film flow model as well as a fuel spray model and an air flow model, are required for the system to represent fuel behavior in intake port and combustion chamber.

Film Flow Model

The newly developed film flow model simulates thin fluid film flow on solid surface of arbitrary configuration. The film thickness is treated as a dependent variable in the model so that the governing equations can be reduced to compact two-dimensional form. Two-dimensional coordinate fitted on arbitrary surface is projected on flat two-dimensional plane to build following governing equations.

Mass conservation equation is,

$$\frac{\partial d}{\partial t} + \frac{1m}{\sin\varphi}\left\{ \frac{\partial}{\partial x}\left(\frac{\sin\varphi}{1m} d\, v_x \right) + \frac{\partial}{\partial y}\left(\frac{\sin\varphi}{1m} d\, v_y \right) \right\} = \frac{\dot{m}_{dep}}{\rho} \tag{1}$$

where the term in right hand side represents arrived mass of fuel spray. Momentum conservation equations in x- and y- direction are,

$$\begin{aligned} \frac{\partial (d\, v_x)}{\partial t} &+ \frac{1m\,B}{\sin\varphi}\frac{\partial}{\partial x}\left(\frac{\sin\varphi}{1m} d\, v_x^2 \right) \\ &+ \frac{1m\,B}{\sin\varphi}\frac{\partial}{\partial y}\left(\frac{\sin\varphi}{1m} d\, v_x v_y \right) \\ &= \frac{p}{\rho}\,\bar{1} + d\, g_x + \frac{1}{\rho}\left(\dot{m}_{dep} v^{\xi}_{dep} + F^{\xi}_{air} - F^{\xi}_{wall} \right) \end{aligned} \tag{2}$$

$$
\frac{\partial (d\, v_y)}{\partial t} + \frac{l\, m\, B}{\sin\varphi}\frac{\partial}{\partial x}\left(\frac{\sin\varphi}{l\, m} d\, v_y\, v_x\right)
+ \frac{l\, m\, B}{\sin\varphi}\frac{\partial}{\partial y}\left(\frac{\sin\varphi}{l\, m} d\, v_y^2\right)
= \frac{\underline{p}}{\rho}\,\overline{m} + d\, g_y + \frac{m}{\rho}\left(\dot{m}_{dep}\, v^{\eta}_{dep} + F^{\eta}_{air} - F^{\eta}_{wall}\right) \tag{3}
$$

where F_{air} and F_{wall} in right hand side represent frictional force on air-fuel and fuel-wall interface, respectively. B is a correction factor for representing velocity distribution in thickness direction by mean velocity defined as,

$$
B = \int_0^d v'^2_{\xi}\, d\,d \,/\, v_{\xi}^2 \tag{4}
$$

For uniform distribution, B = 1, and for linear distribution, B = 4/3. Here, B of 1.125 is employed under assumption of gravity dominant field. Momentum conservation equation in z-direction yields,

$$
\frac{\underline{p}}{\rho}\left(\overline{n} - \overline{l}\,\frac{\sqrt{1-l^2}}{l} - \overline{m}\,\frac{\sqrt{1-m^2}}{m}\right)
= B\, d\, v_x^2 \frac{\partial}{\partial x}\frac{\sqrt{1-l^2}}{l} + B\, d\, v_y^2 \frac{\partial}{\partial y}\frac{\sqrt{1-m^2}}{m} +
B\, d\, v_x\, v_y\left(2\frac{\partial^2 z}{\partial x\,\partial y}\right) + \frac{\sqrt{1-l^2}}{l} d\, g_x + \frac{\sqrt{1-m^2}}{m} d\, g_y \tag{5}
$$

Further details on derivation of the equations are described in reference (5).

This new model is verified by an experiment of simple film flow on a flat plane. Good agreement of experimental and calculated mass flow distribution is obtained, which suggested the validity of the model. Detail of the verification is shown in Appendix A and also in reference (5).

Fuel Spray Model

Fuel spray is simulated by Discrete Droplet Model (DDM) which treats spray as numbers of parcels containing droplets of the same attributes. Motion of a parcel is described as,

$$
\left(\frac{d V_d}{d t}\right)_j = \frac{3}{4} C_d \frac{\rho_{air}}{\rho}\frac{1}{d_{drop}}\left(V_a - V_d\right)\left|V_a - V_d\right| \tag{6}
$$

$$
j = x, y, z
$$

$$
C_d = \left(0.63 + \frac{4.8}{\sqrt{Re_d}}\right)^2 \tag{7}
$$

$$
Re_d = \frac{d_{drop}\left|v_a - v_d\right|}{\nu} \tag{8}
$$

which are similar as in reference (2). In reference (2) and (3), complex phenomena of droplets, such as breakup or collision, are modeled to predict detailed behavior of droplets in diesel spray. In this study, those sub-models are not included because those phenomena are not supposed to be critical in gasoline spray. Also, evaporation of spray is not included in this study, because the cold engine condition is supposed important regarding with the fuel film flow.

Air Flow Model

In this study, commercial package of air flow model is employed to calculate three-dimensional air flow field in

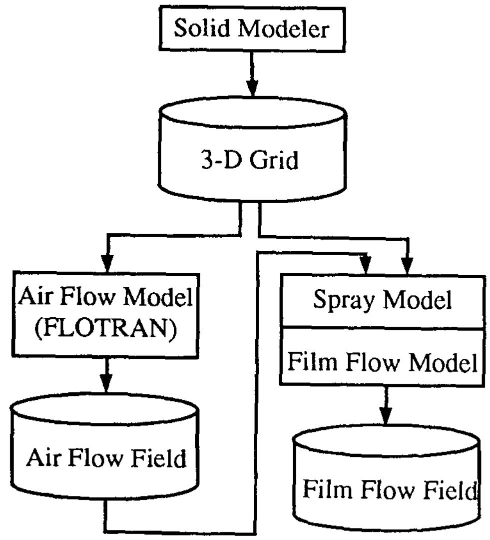


Fig. 1 Data Flow in the System

intake port (6). The solver is finite element method type. Flow is assumed steady, compressible, turbulent flow.

System

Those three models are combined to build a system which simulates fuel behavior in intake port and combustion chamber. Schematic diagram of data flow in the system is shown in Fig. 1.

In this study, as the commercial package is employed for air flow solver which cannot be modified to interact with the other models, air flow is calculated solely preliminary, then, spray and film flow are calculated according to the air flow field. Therefore, the air flow field gives effects on spray and film flow, but not vice versa.

When droplets of spray reach and impinge on a wall, “splash” may occur. This splash phenomena should be included in the system, however, the phenomena is so complicated that observation studies are on their way and compact models suitable for this study are not available. Therefore, in this study, employing an assumption that splash occurs in the same manner, a droplet arriving on wall is treated such that fixed rate of mass of the droplet gives mass and momentum to form or join the film flow, and the rest of the droplet reflects and fly away. Here, 0.2 is employed as the rate of reflection.

Computational Mesh

Computational mesh geometry used in following study is shown in Fig. 2. The mesh is unstructured, tetrahedron type, whose number of elements is 45968, and nodes, 9306. The whole elements are used to calculate air flow field and droplets trajectory, while in the film flow calculation, two-dimensional triangular elements on solid surface, extracted from the three-dimensional elements, are used.

Fig. 2 Mesh Geometry

DISCUSSION

Verification of the system were undertaken by following comparison of calculated results with those from experiments.

A cylinder head of a conventional gasoline engine, with fixed intake valve lift, was installed on steady flow test rig, and steady air flow was induced in the intake port by sucking air from the bottom of the cylinder. Experimental condition is shown in Table 1. Fuel injector was installed in conventional way at entrance section of the intake port as the axis of the spray targeting on the valve head. Condition of the spray is shown in Table 2. Size, velocity, and mass distribution of droplets in the spray were measured by Phase Doppler Particle Analyzer preliminary to use in the calculation as initial conditions of DDM. Distribution of droplet size, measured as well as used in calculation, is shown in Fig. 3.

<u>Circumference Distribution of Fuel Mass Flow at Valve Seat</u>

In the first, circumference distribution of fuel mass flow passing at valve seat from intake port into combustion chamber was measured. Detail of the experiment is described in Appendix B.

Comparison of the fuel mass flow between experimental and calculational result is shown in Fig. 4. General pattern of one peak in (-y) side is well represented. However, circumferential position of the peak does not well agree. The peak position by the experiment locates on more outer direction (+x) than the calculated one. One reason for this discrepancy may be the splash phenomena. In the real fuel spray, droplets impinged on the valve head may reflect and rettach on further port wall, and form the peak at outer direction. In this splashing, the rate of reflection may vary largely from droplet to droplet according to their size, incoming velocity, impinging angle, film condition, or other factors. In the calculation, splash is treated by a simple model such that fixed rate of impinging droplet always reflects. This difference may cause the discrepancy. Also, mass distribution in (-x +y) direction in

Table 1 General Condition of Computation /Experiment

Type of Combustion Chamber	4-valved, pentroof
Bore x Stroke	82.5 x 86.0 mm
Diameter of Intake Valve	32.2 mm
Intake Valve Lift	10 mm (fixed)
Air Flow Rate	4.0×10^{-3} m^3 / s

Table 2 Condition of Spray

Mode of Injection	Continuous intermittent
Cycle of Injection	10 Hz
Duration of Injection	11 ms
Mass of Injection	29 mg / shot (total)
Spray Type	Full Cone
Initial Speed	14 m / s
Sauter Mean Diameter	2.7×10^{-4} m
Cone Anagle	4.2° (half value)

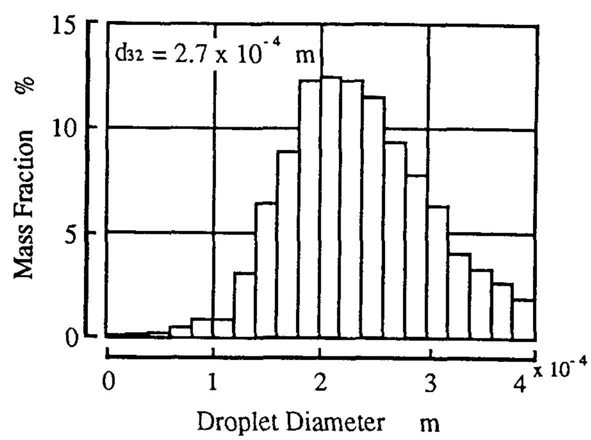


Fig. 3 Droplet Diameter Distribution of the Spray

the experiment may occur by the splash of the droplets impinged on the other side of the valve head.

<u>Fuel Spread on Combustion Chamber</u>

In the second, temporal behavior of fuel spread on upper wall of combustion chamber just after beginning of injection was observed. Laser was lit on the combustion chamber wall to induce luminescence light from the fuel, and pictures were taken to compare with those by calculation.

Experimental pictures of fuel film flow behavior from the beginning of injection to 1.8 seconds after that are shown in Fig. 5. Here, white parts captured in the pictures are luminescent light induced by laser, which indicates

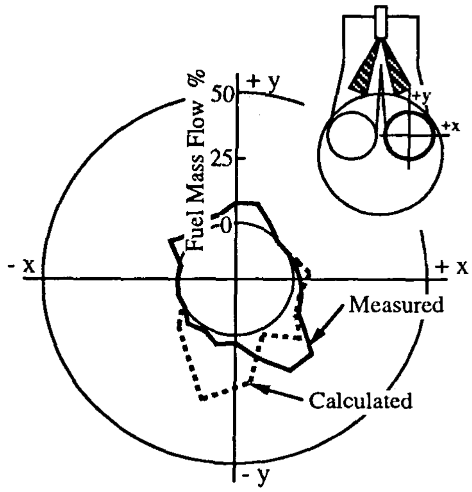


Fig. 4 Circumference Distribution of Fuel Mass Flow at Valve Seat

existence of fuel. Thickness of the fuel film is not well resolved. Calculated fuel film flow behavior for the same condition as Fig. 5 is shown in Fig. 6. Comparing Fig. 6 with Fig. 5, the calculation shows good agreement on behavior of the film flow spreading out on the upper wall of the combustion chamber.

In summary for the comparison above, this system has been found possessing ability of qualitative prediction of fuel film behavior in intake port and combustion chamber of gasoline engines.

APPLICATION TO ENGINE DESIGN

Using this system, four cases of fuel spray design, shown in Table 3, are studied in calculation. CASE-STD is the basecase, whose condition is the same as that in above discussion. Comparing with CASE-STD, CASE-A has wider cone angle, CASE-B, small Sauter mean diameter, and CASE-C, both.

Table 3 Spray Parameter of Case Study

CASE	Cone Angle °	Sauter Mean Diameter m	Other Condition
CASE-STD	4.2	2.7×10^{-4}	same as Table 2
CASE-A	7.0	2.7×10^{-4}	
CASE-B	4.2	5.0×10^{-5}	
CASE-C	7.0	5.0×10^{-5}	

CASE-STD

Spray behavior of CASE-STD is shown in Fig. 7. The calculation was for one shot of injection, from the beginning of injection to 40 ms after that, supposing most of the droplets finish their movement in this time of duration. Most of the droplets fly straightly as injected, which suggests the droplets size are large enough not to be affected by the air flow. Most of the droplets impinge on the valve stem, the valve head, and lower part of the intake port upper wall, and rests go through clearance between the valve and the valve seat to impinge on the cylinder wall.

Film flow behavior formed on the intake port and upper wall of the combustion chamber from the beginning of injection to 2.0 s after that is shown in Fig. 8. Thickest part of the film is formed around the valve seat position; this is because the impinging position of the droplets that impinge on outer wall concentrates on the valve seat position. Then, this thick film begins to spread out on upper wall of the combustion chamber, and flow down the side wall.

CASE-A (wide cone angle)

Spray behavior of CASE-A is shown in Fig. 9, and film behavior in Fig. 10. Those behavior are almost the same as those of CASE-STD in Fig. 7 and 8. Major observable difference is width of the thickest part of the film formed around valve seat position. CASE-A has wider area in circumference direction than in CASE-STD. Also, CASE-A has more droplets flying through clearance between the valve and the valve seat into the combustion chamber and impinge on the cylinder wall. This calculation suggests possibility of forming more film flow on port wall when trying to target the spray on valve clearance to put more droplet directly into cylinder.

CASE-B (small Sauter mean diameter)

Spray behavior of CASE-B is shown in Fig. 11. Droplets with smaller diameter get more dominant effect by air flow because momentum get relatively smaller than air drag force. Therefore, in CASE-B, trajectory of droplets are more largely bent by the air flow than CASE-STD in Fig. 7, and most of the droplets impinge and attach on upper wall of the port. Behavior of film flow, which is formed by those attaching droplets and creeps down along the port into upper wall of the combustion chamber, is shown in Fig. 12. Fuel spray of smaller diameter has advantage of easier evaporation, better fuel-air mixing, and better combustion. However, this calculation showed possibility of forming more film flow on the port in case fuel is injected in a intake process, which can lead to worse combustion or more response delay, unless special care is taken.

CASE-C (wide cone angle, small Sauter mean diameter)

Spray and film flow behavior of CASE-C, combination of CASE-A and CASE-B, are shown in Fig. 13 and 14. Those behavior are almost the same as CASE-B such that most of the droplets impinge on upper wall of the port and form film flow creeping down the port into upper wall of the combustion chamber. This calculation indicates that droplet size effect is dominant comparing with cone angle effect for this cases. Though if much smaller cone angle is employed, the behavior of spray and film flow is not expected to change largely as far as the initial position of the spray is kept the same. Therefore, when employing spray of smaller diameter, special care should be taken on injector position rather than other injector characteristics.

In summary of those case studies, this system has been found being a useful tool in engine designing, especially on conceptual design stages. The system can provide schematic and positive figures of fuel behavior, which lead to deeper understanding of phenomena and better ideas to alternative designs.

Some of further applications of the system on engine design will be following.

- Estimation of “fuel spray impingement rate on intake valve” under steady flow condition. This rate is an index to measure how much of the injected fuel spray directly flies into combustion chamber, used in conceptual design stage. Up to the present, estimation of this rate is based on only geometry of intake port, valve, and spray configuration, which does not well represent the rate in real engine condition where air flow is there bending the spray. Using this system, spray behavior being affected by air flow can be predicted, and the rate of impingement on the intake valve will be more realistic.

- Determination of spark plug position. Fumigation of spark plug, due to fuel film flow on upper wall of combustion chamber wetting the plug, causes poor start-up performance in cold engine condition and / or much hydrocarbon emission. Using the film flow behavior on combustion chamber predicted by this system, less wet position on upper wall of the combustion chamber can be determined to install spark plug. Also, intake port and spray layout of forming less film flow can be examined.

CONCLUSION

A newly developed film flow model was combined with a fuel spray model and an air flow model to analyze fuel film behavior in intake port and combustion chamber of gasoline engines.

The calculated results of the system were compared with experiment, and the system has been found possessing ability of qualitative prediction of the fuel film behavior.

Applying on case studies, this system has been found being a useful tool in engine design stages, especially on conceptual designing stages.

Future study of this system should be emphasized in following fields.

Suitable splash model should be developed and built in the system. Splash phenomena is treated by a simple model in this study so that discrepancy of fuel distribution may have occurred. Splash should be modeled more precisely, at the same time compactly enough to build in this system, to give less discrepancy.

Also, the system should be extend to be capable of calculation under unsteady condition. In this study, air flow and fuel injection is at steady condition. Those results from calculation of steady condition, which were not available before, provide useful information on designing stage. However, calculation under unsteady condition, such that piston and valve motion as well as injection timing are included, will give much more information.

NOMENCLATURE

B	correction factor defined by Eqn. (4)
C_d	drag coefficient defined by Eqn. (7)
d	film thickness
d_{drop}	diameter of droplet
F^{ξ}_{air} , F^{η}_{air}	ξ- and η-directional component of frictional force at the interface between film and air
F^{ξ}_{wall} , F^{η}_{wall}	ξ- and η-directional component of frictional force at the interface between film and wall
g_x, g_y	x- and y-directional component of acceleration
l, m, n	directional cosine of the mesh
m_{dep}	mass of droplets arrived on the mesh per unit area per unit time
p	pressure
Re_d	Reynolds number difinecd by Eqn. (8)
t	time
V_x, V_y	x- and y-directional component of droplet velocity
V^{ξ}_{dep} , V^{η}_{dep}	ξ and η-directional component of velocity of droplet arrived on the mesh
V_a	velocity of air
V_d	velocity of droplet
x, y	coordinate in projected plane
ξ, η	coordinate on arbitrary surface in real space
ν	dynamic viscosity of air
ρ	density of the fluid
ρ_{air}	density of air
φ	angle between ξ and η on mesh

Reference

(1) K. Naitoh, H. Fujii, T. Urushihara, and Y. Takagi, “Numerical Simulation of the Detailed Flow in Engine Ports and Cylinders”, SAE 900256

(2) A.A. Amsden, P. J. O’Rourke, and T. D. Butler, “KIVA-II: A Computer Program for Chemically Reactive Flows with Sprays”, Los Alamos National Laboratory Report LA-11560-MS

(3) J. D. Naber and R. D. Reitz, “Modeling Engine Spray / Wall Impingement”, SAE 880107

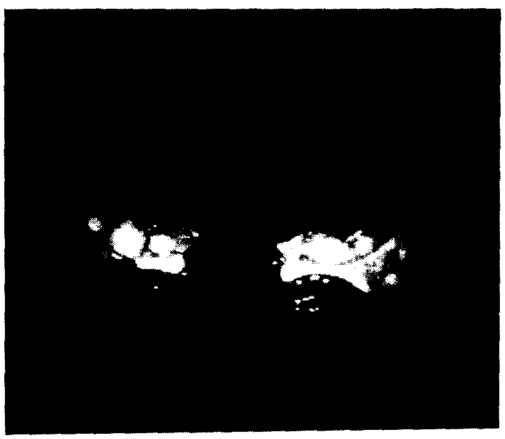
(4) K. Takamori, R. Kawabe, T, Kawasaki, M. Naitou, and T. Takamaru, “Numerical Simulation of the Fuel Flow in an Automobile Fuel Supplier (4th Report, Three-Dimensional Numerical Analysis of Fuel Flow in an Intake Manifold), Transaction of the Japan Society of Mechanical Engineers, 56, 528, (in Japanese)

(5) T. Kawasaki, et. al. “Development of analytical Method for Fuel film Flow in Spark Ignition Engine”, Japan Society of Mechanical Engineers International Journal, (to be published)

(6) FLOTRAN Manual, Compuflo, Inc.

Time = 0.2 s after Injection Began

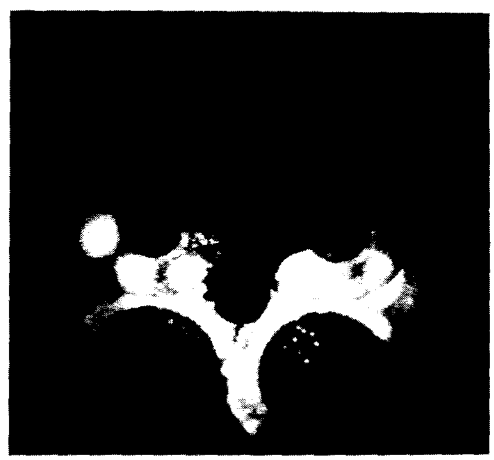
Time = 0.5 s



Time = 0.8 s

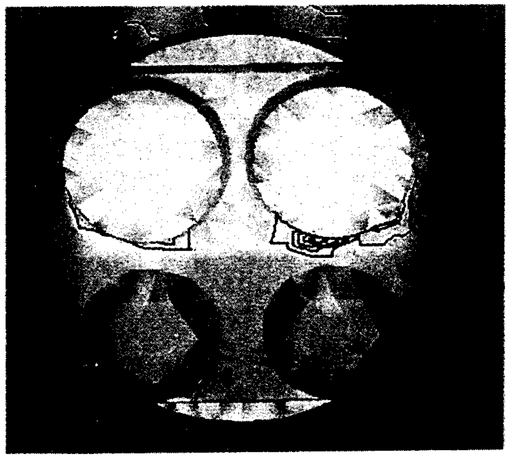
Time = 1.2 s

Time = 1.5 s

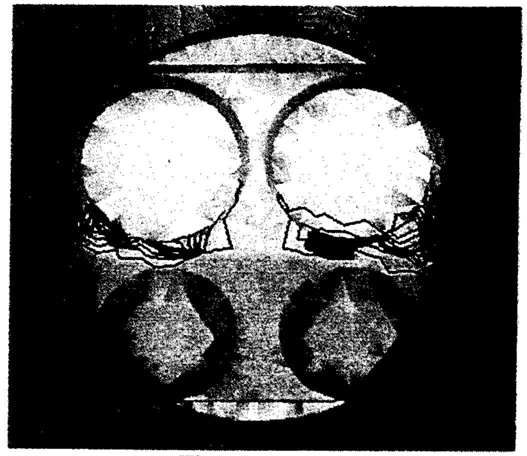


Time = 1.8 s

Fig. 5 Film Flow Behavior on Combustion Chamber (Experiment)



Time = 0.2 s after Injection Began



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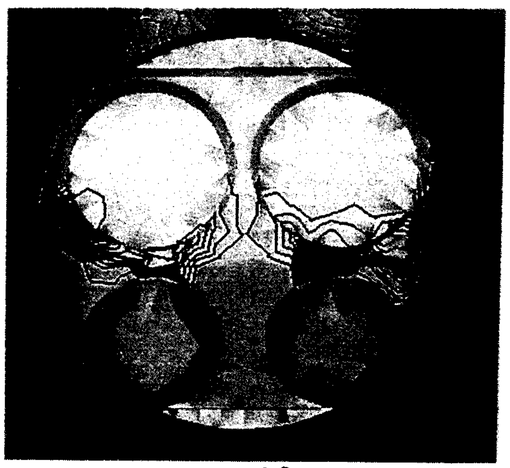

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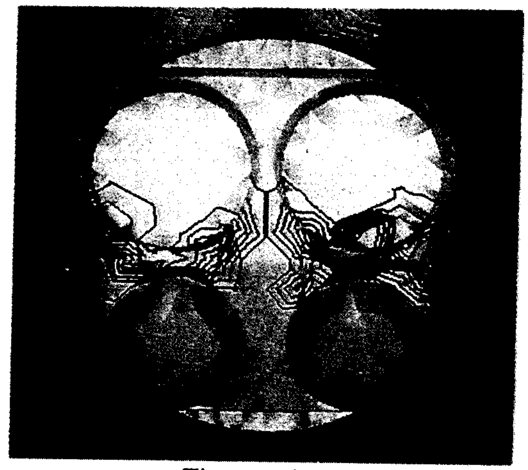

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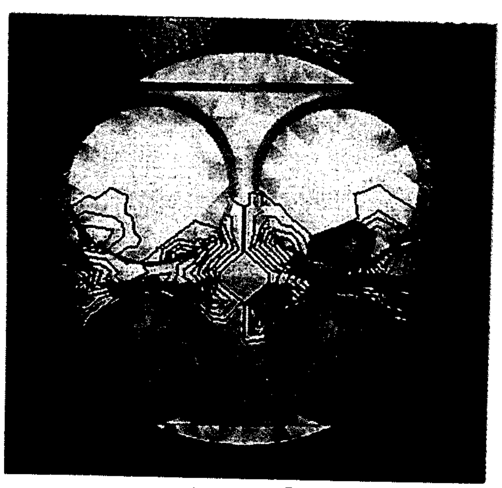

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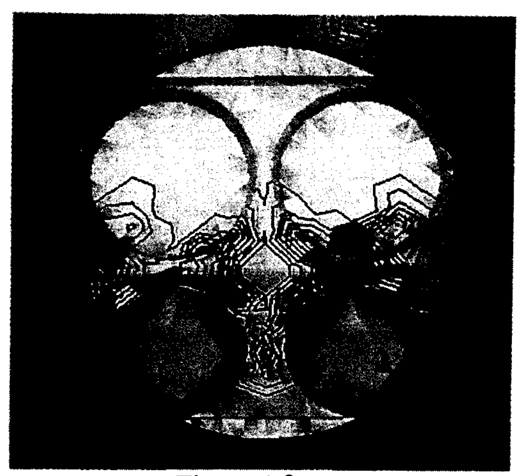
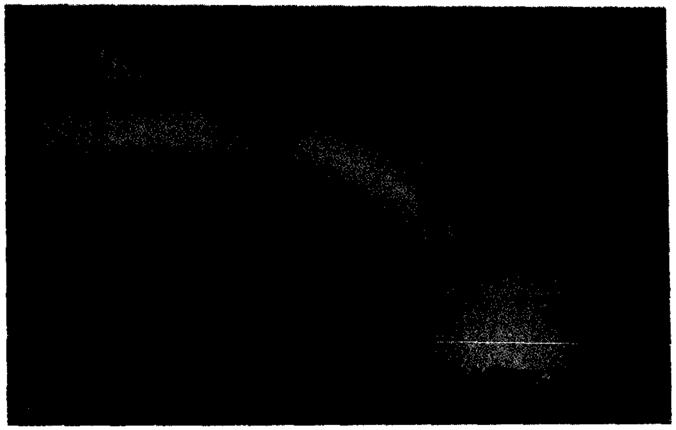


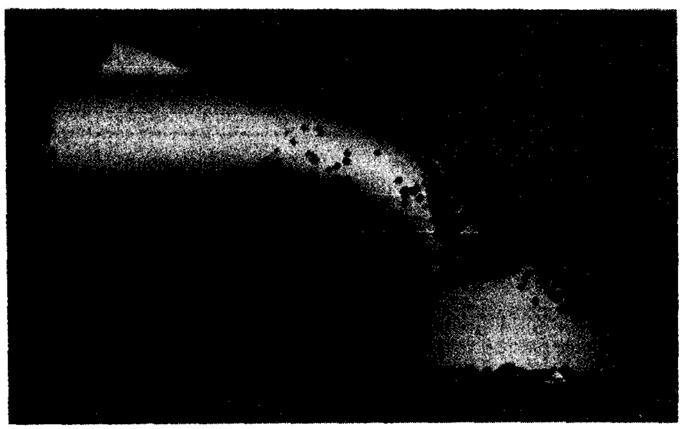
Time = 1.8 s

Fig. 6 Film Flow Behavior on Combustion Chamber (Calculation)



Time = 5 ms after injection began

Time = 10 ms



Time = 20 ms

Time = 40 ms

Fig. 7 Spray Behavior (CASE-STD)

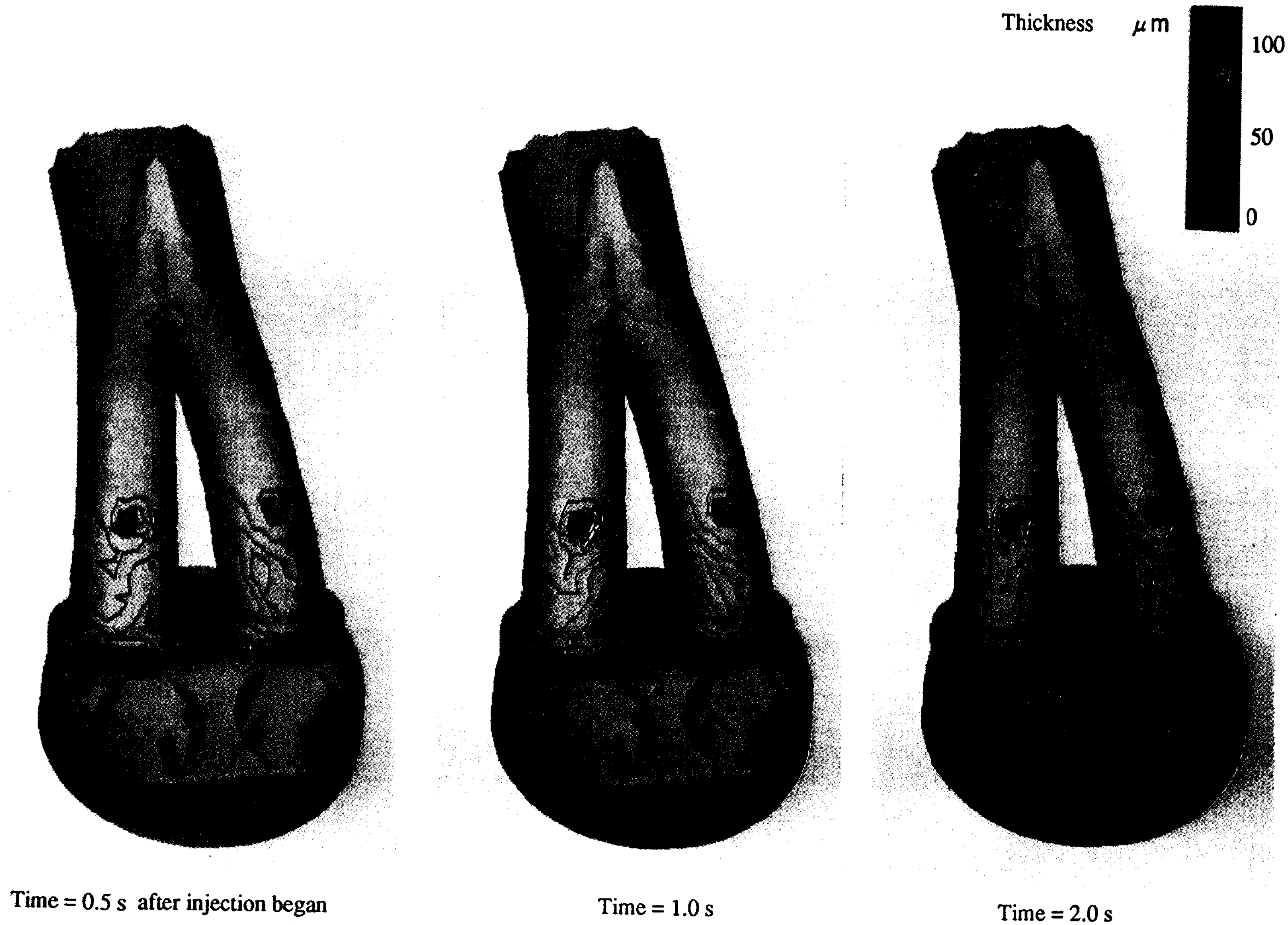
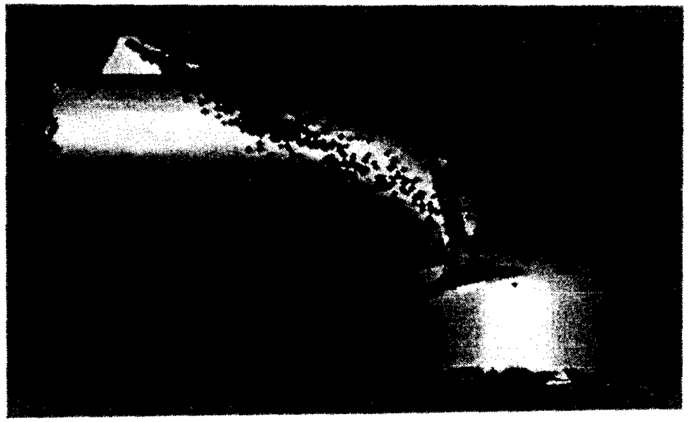


Fig. 8 Film Flow Behavior on Intake Port Wall (CASE-STD)



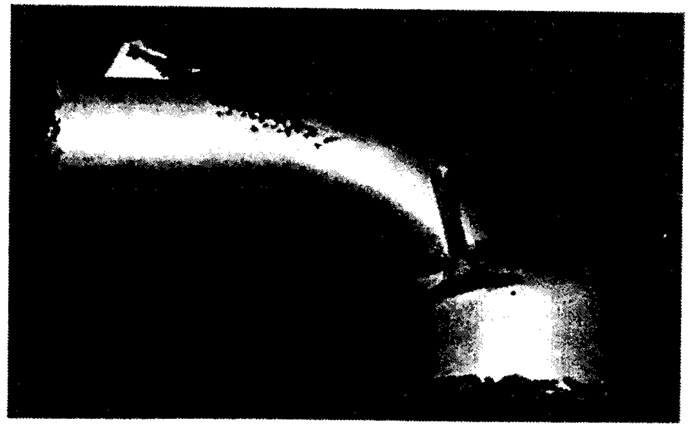
Time = 10 ms after injection began

Time = 40 ms

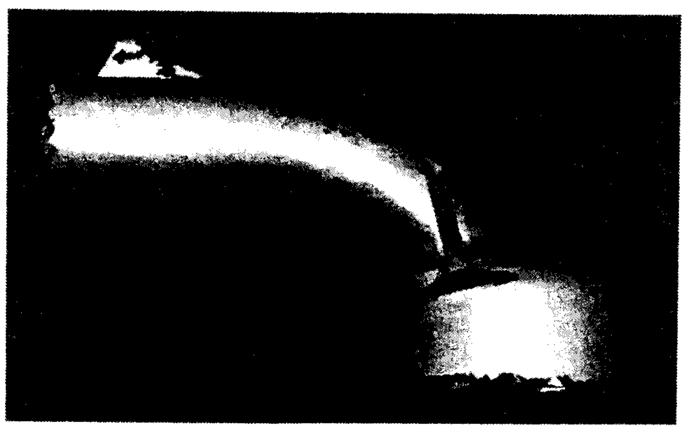
Fig. 9 Spray Behavior (CASE-A)

Time = 0.5 s after injection began

Fig. 10 Film Flow Behavior on Intake Port Wall (CASE-A)

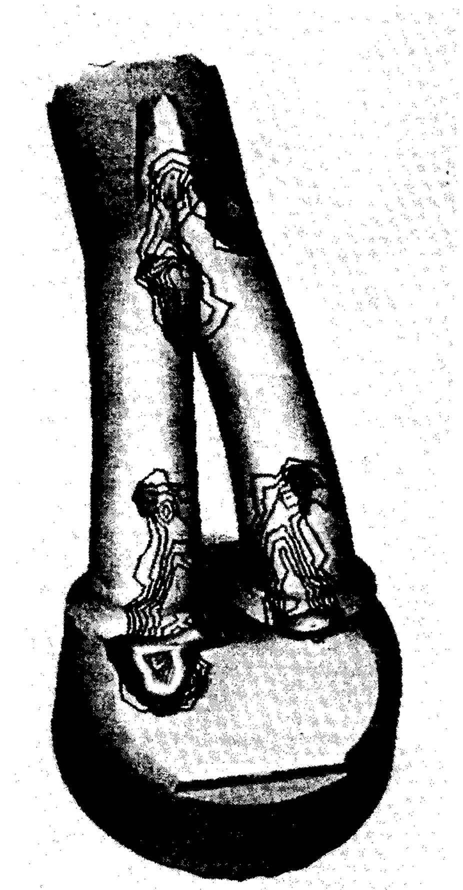


Time = 10 ms after injection began



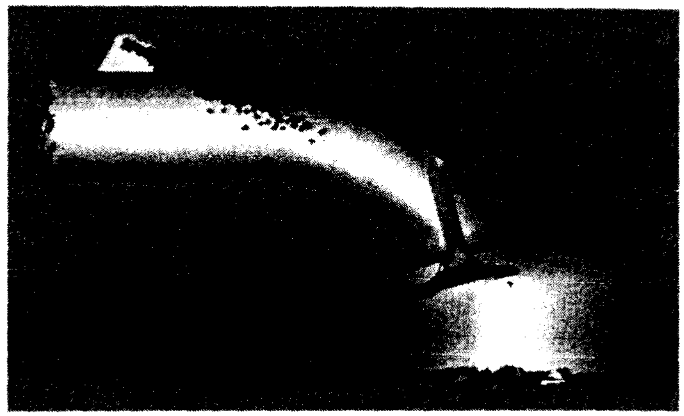
Time = 40 ms

Fig. 11 Spray Behavior (CASE-B)

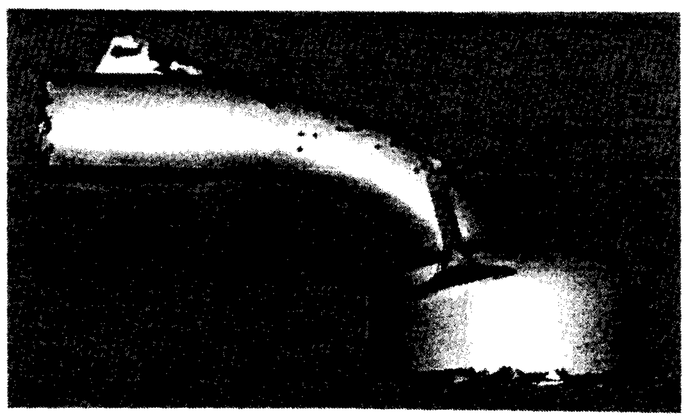


Time = 0.5 s after injection began

Fig. 12 Film Flow Behavior on Intake Port Wall (CASE-B)



Time = 10 ms after injection began



Time = 40 ms

Fig. 13 Spray Behavior (CASE-C)

Time = 0.5 s after injection began

Fig. 14 Film Flow Behavior on Intake Port Wall (CASE-C)

APPENDIX A Verification of Film Flow Model

Film flow model was verified by experiment in one of the simplest condition. A flat plate is installed inclined, and fuel is injected continuously to form film flow creeping down the plane. At the lower edge of the plate, mass distribution of the film flow is measured using a small gate connecting to reservoir. Schematic diagram of the experiment is shown in Fig. A1. Comparison between the experiment and calculation results is shown in Fig. A2. The mass flow distribution calculated by the film flow model shows good agreement with those by experiment, which suggests the validity of the model. In this comparison, rate of reflection is also studied, and 0.2 is found to give good agreement.. Further detail is also described in reference (5).

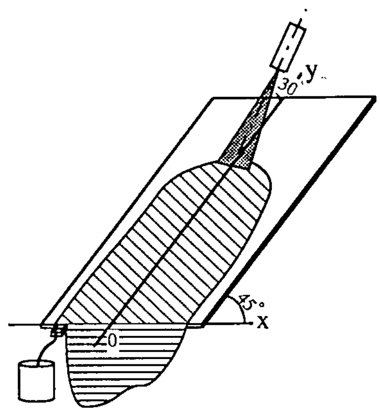


Fig. A1 Layout of Simple Flow Experiment

Table A1 Condition of Spray

Mode of Injection	Continuous intermittent
Cycle of Injection	10 Hz
Duration of Injection	11 ms
Mass of Injection	20 mg / shot
Spray Type	Full Cone
Initial Speed	11 m / s
Sauter Mean Diameter	1.9×10^{-4} m
Cone Anagle	4.2° (half value)

APPENDIX B Experimental Method to Measure Circumference Mass Flow Distribution at Valve Seat

The valve seat employed in the experiment to measure circumference mass flow distribution at valve seat position is shown in Fig. B1. 24 pipes of radius 1.0 mm were installed on circumference of the valve seat in equi-distance. In the measurement, this valve seat was inserted on the cylinder head in Fig. 3, and one of the pipe was connected to suction pump via a reservoir tank. When fuel was injected in steady manner, fuel film formed on the port wall flowed down into the combustion chamber passing the valve seat. Then, fuel film passing at the pipe position

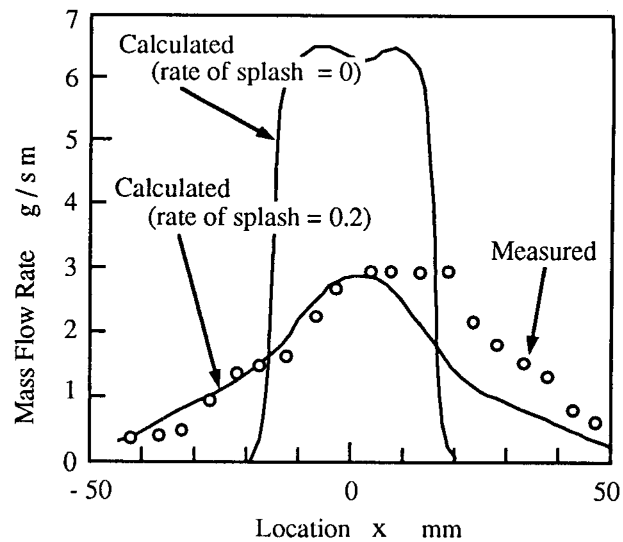


Fig A2 Mass Flow Distribution of Film Flow on Inclined Flat Plate

was sucked into the pipe and accumulated in the reservoir tank. After a certain fixed time, enough mass was accumulated in the tank. In the same manner, the induction pump was reconnected to another pipe one by one to measure whole distribution of mass flow in circumference direction.

Here, each pipe is suppose to suck the fuel film on the one twenty-fourth of the valve area around the pipe. However, “crosstalk” may occur, in which one pipe may cover more than the one twenty-fourth area and collect part of mass allocated to the adjacent pipes. Though this crosstalk is supposed to be small, the total mass may become larger than it should be. Therefore, to cancel this crosstalk effect, the measured mass are normalized after measuring the whole pipes. The calculate result is also normalized to compare with them.

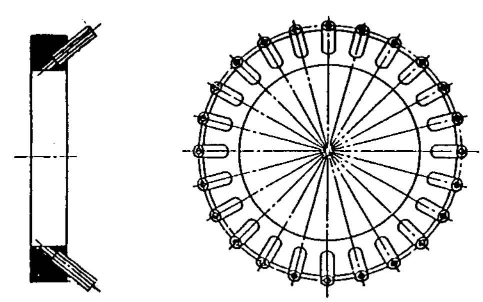


Fig. B1 Valve seat for Measurement of Circumference Mass Flow Distribution at Valve Seat Position